In-Class Activity: PK-PD

Module 7b - PK-PD Relationship · In-class activity · 12 Oct 2026

Name _____ Date _____

1. A drug with an elimination half-life of 3 hours, apparent volume of distribution of 25 L, and minimum effective concentration of 1.2 mcg/mL is to be administered as a single 250 mg IV bolus dose.

1a What is the duration of action of this dose?

1b What dose would yield a duration of action of 12 hours?

2. The equation for concentration of drug in the plasma as a function of time following

a 500 mg IV bolus dose was found to be:

2a $C_p = 50e^{-0.2t}$ a. What is the duration of action of a single 500 mg IV bolus dose if the minimum effective concentration is 5 mcg/mL?

2b $C_p = 50e^{-0.2t}$ b. What dose would double the duration of action found in part a?

2c $C_p = 50e^{-0.2t}$ c. What is the expected change in the duration of action following a single 250 mg IV bolus dose instead of the 500 mg IV bolus dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 9.16 hr

WHAT THE STEM GIVES

t½	3 hours	multiplied by 1.44 outside the logarithm
V_D	25 L	times Ceff gives 30 mg
Ceff	1.2 mcg/mL	equals 1.2 mg/L, the MEC
D_0	250 mg	the numerator inside the logarithm

THE WORKING, ONE LINE AT A TIME

UNIT

$$\text{Ceff} = 1.2 \text{ mcg/mL} = 1.2 \text{ mg/L}$$

1 mcg/mL equals 1 mg/L, because multiplying top and bottom by 1000 turns mcg into mg and mL into L. With the dose in mg and the volume in L, the ratio inside the logarithm has no units.

SETUP

$$\text{teff} = 1.44\ t_{½}\ \ln\left(\frac{D_0}{V_D \times \text{Ceff}}\right)$$

The stem gives the half-life, not k, so the half-life form of the duration line applies directly; 1.44 is 1 ÷ 0.693, the step from k to t½.

ALGEBRA

$$\text{teff} = 1.44\ (3 \text{ hr})\ \ln\left(\frac{250 \text{ mg}}{(25 \text{ L})(1.2 \text{ mg/L})}\right) = 4.32 \text{ hr} \times \ln\left(\frac{250}{30}\right) = 4.32 \text{ hr} \times \ln 8.333 = 4.32 \text{ hr} \times 2.120$$

$V_D \times \text{Ceff} = 30$ mg is the amount in the body when the level is at the MEC. The dose is 8.33 times that, and ln 8.333 = 2.120.

ROUND

$$\text{teff} = 9.16 \text{ hr}$$

4.32 × 2.120 = 9.16 hr. With k = 0.231 hr^{-1} in place of 1.44 t½ the result is 9.18 hr; the 1.44 is rounded, so the two forms differ only in the second decimal.

> **DOES THE NUMBER MAKE SENSE?**
> 250 mg is 8.333 times the 30 mg at the MEC, between 2 to the 3rd and 2 to the 4th, so teff is 3 to 4 half-lives of 3 hr: 9.16 hr.

Her working gives 9.15; the 1.44 t½ form gives 9.16 hr and k = 0.231 hr^{-1} gives 9.18 hr. All three are within the margin.

1b

Worked by the drill

Answer: 479.7 mg

WHAT THE STEM GIVES

teff	12 hours	the target duration, in the exponent
t½	3 hours	gives k = 0.231 hr^{-1}
V_D	25 L	times Ceff gives 30 mg
Ceff	1.2 mcg/mL	equals 1.2 mg/L, the MEC

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{3\ hr} = 0.231\ hr^{-1};\ Ceff = 1.2\ mcg/mL = 1.2\ mg/L$$

The dose line has k in its exponent, so the 3-hour half-life is turned into k first. 1 mcg/mL equals 1 mg/L, so the MEC is 1.2 mg/L.

ALGEBRA

$$Ceff = \frac{D_0}{V_D}\ e^{-k\ teff},\ \text{so}\ D_0 = Ceff \times V_D \times e^{k\ teff}$$

$C = C_0e^{-kt}$ with $C_0 = D_0$ over V_D, written at the moment the level reaches the MEC, 12 hours after the dose, then solved for the dose.

ALGEBRA

$$D_0 = (1.2\ mg/L)(25\ L)\ e^{(0.231)(12)} = 30\ mg \times e^{2.772} = 30\ mg \times 15.99 = 479.7\ mg$$

mg/L × L leaves mg: 30 mg is the amount in the body at the MEC. kt = 0.231 hr^{-1} × 12 hr = 2.772 has no units, and the dose must be $e^{2.772}$ = 15.99 times that 30 mg.

ROUND

$$D_0 = 479.7\ mg$$

Her value is 479.72 mg. 12 hours is 4 half-lives of 3 hr, and 2 to the 4th is 16, so the dose is about 16 times the 30 mg at the MEC.

> **DOES THE NUMBER MAKE SENSE?**
> 12 hours is 4 half-lives of 3 hr, so the dose is about 2 to the 4th, 16, times the 30 mg at the MEC: 479.7 mg, above the 250 mg of part a.

2a

Worked by the drill

Answer: 11.51 hr

WHAT THE STEM GIVES

C_0	50	mg/L, the coefficient of the equation
k	0.2t	k = 0.2 hr^{-1}, the exponent
D_0	500 mg	over C_0 gives V_D = 10 L
Ceff	5 mcg/mL	equals 5 mg/L, the MEC

THE WORKING, ONE LINE AT A TIME

SETUP

From $C_p = 50e^{-0.2t}$: $C_0 = 50$ mg/L and $k = 0.2\ hr^{-1}$; $V_D = \frac{500\ mg}{50\ mg/L} = 10\ L$

An IV bolus equation reads as $C = C_0e^{-kt}$: the number in front is C_0 and the exponent carries k. V_D is the dose over C_0.

UNIT

Ceff = 5 mcg/mL = 5 mg/L

1 mcg/mL equals 1 mg/L, because multiplying top and bottom by 1000 turns mcg into mg and mL into L; the MEC can then be set against C_0 = 50 mg/L directly.

ALGEBRA

$$t_{eff} = \frac{\ln C_0 - \ln Ceff}{k} = \frac{\ln(50 \div 5)}{0.2\ hr^{-1}} = \frac{\ln 10}{0.2} = \frac{2.303}{0.2\ hr^{-1}} = 11.51\ hr$$

The level has to fall tenfold, from 50 to 5 mg/L. With k read off the equation the natural-log form needs no half-life.

ROUND

$$t_{½} = \frac{0.693}{0.2} = 3.465\ hr;\ t_{eff} = 11.51\ hr$$

The half-life form gives 1.44 (3.465 hr) ln 10 = 11.49 hr, because 1.44 is rounded from 1 ÷ 0.693; both are the same answer within the margin.

DOES THE NUMBER MAKE SENSE?

The level must fall tenfold, between 2 to the 3rd and 2 to the 4th, so teff is 3 to 4 half-lives of 3.465 hr: 11.51 hr.

Her working stops at $t_{½}$ = 3.465 hr, V_D = 10 L and the 1.44 $t_{½}$ set-up; this answer is worked from her stem by that set-up.

2b

Worked by the drill

Answer: 5000 mg

WHAT THE STEM GIVES

teff	11.51 hr	doubled to 23.03 hr, the target
Ceff	5 mcg/mL	equals 5 mg/L
C_0	50	mg/L; 500 mg over 50 mg/L gives V_D = 10 L
k	0.2t	k = 0.2 hr^{-1}, in the exponent

THE WORKING, ONE LINE AT A TIME

SETUP

Target $t_{eff} = 2 \times 11.513\ hr = 23.03\ hr$; $V_D = \frac{500\ mg}{50\ mg/L} = 10\ L$; $k = 0.2\ hr^{-1}$

Doubling the duration means the level must still be at the MEC 23.03 hours after the dose; part a unrounded is 11.513 hr. V_D and k come from the equation and do not change with the dose.

ALGEBRA

$D_0 = Ceff \times V_D \times e^{k\ teff} = (5\ mg/L)(10\ L)\ e^{(0.2)(23.026)} = 50\ mg \times e^{4.605}$

$C = C_0 e^{-kt}$ solved for the dose at the moment the level reaches Ceff. 5 mcg/mL is 5 mg/L, and mg/L × L leaves 50 mg, the amount in the body at the MEC.

ALGEBRA

$e^{4.605} = 100$, so $D_0 = 50\ mg \times 100 = 5000\ mg$

At 500 mg the level falls tenfold over 11.51 hr; over twice the time it must fall tenfold twice, a hundredfold, so C_0 must be 100 × 5 = 500 mg/L.

ROUND

$D_0 = 5000$ mg (5 g), ten times the 500 mg dose

Doubling the duration took ten times the dose, not twice it, because the duration grows with the logarithm of the dose.

> **DOES THE NUMBER MAKE SENSE?**
> Twice the duration needs the tenfold fall twice over, a hundredfold, so the dose is 100 times the 50 mg at the MEC: 5000 mg, ten times 500 mg.

Left unworked on the activity sheet; this answer is worked from her stem by the dose line she uses in problem 1b.

2c

Worked by the drill

Answer: 3.47 hr shorter

WHAT THE STEM GIVES

D_0	250 mg	half of 500 mg, so one doubling less
k	0.2t	k = 0.2 hr^{-1}, so t½ = 3.47 hr
Ceff	5 mcg/mL	equals 5 mg/L
teff	11.51 hr	the 500 mg duration, compared against

THE WORKING, ONE LINE AT A TIME

SETUP

C_0 at 250 mg $= \dfrac{250\ mg}{10\ L} = 25$ mg/L; teff $= \dfrac{\ln(25 \div 5)}{0.2\ hr^{-1}} = \dfrac{1.6094}{0.2} = 8.047$ hr

V_D = 10 L from part a does not change with the dose, so half the dose gives half of C_0, 25 mg/L. The level then has to fall only fivefold to reach the MEC of 5 mg/L.

ALGEBRA

Change = 8.047 hr − 11.513 hr = −3.466 hr

Halving the dose takes away one doubling, and each doubling adds one half-life of duration: t½ $= \dfrac{0.693}{0.2\ hr^{-1}} = 3.47$ hr.

ROUND

The duration is 3.47 hr shorter, one half-life

−3.466 hr is 3.47 hr shorter to two decimal places: a decrease of one half-life, whatever the starting duration; it is not half of 11.51 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Half the dose removes one half-life, 3.466 hr, from 11.513 hr, leaving 8.047 hr: shorter, but not halved.

Keyed as the size of the change; the duration falls, from 11.51 hr to 8.05 hr. Left unworked on the activity sheet; worked from her stem.
